SUDARSAN SRINIVASAN

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SUMMARY

- Product Design Engineer with 4+ years spanning consumer product development and automotive powertrain design, with a track record of shipping hardware at high volume
- Experienced across the full development cycle from concept and detailed design through validation and production release, working across industrial design, manufacturing, and supply chain teams
- Skilled in DFM/DFA, GD&T, tolerance analysis, FEA, and failure analysis across plastic, metal, and die cast components
- Available for full-time roles starting September 2026

EXPERIENCE

SharkNinja ***Jan 2026 - Present***

Product Design Engineering Co-op, New Product Development *Needham, MA*

- Identified a design flaw in a Slushi machine component, proposed a design change, prototyped the fix, and validated through testing to resolve the issue
- Devised a condensation validation approach using fine mist spray to simulate surface moisture buildup, replacing a humidity chamber setup and cutting test cycle time by ~80%
- Iterated on prototype builds across hardware and software versions, benchmarking performance against KPIs and recommending design or algorithm adjustments
- Collaborated across test kitchen, controls, and advanced development teams, coordinating with counterparts in China to align testing and development efforts

The Boring Company ***Sep 2025 - Dec 2025***

Integration Engineering Intern *Bastrop, TX*

- Bridged design to manufacturing for the MB-1 tunnel boring machine by reviewing CAD assemblies for functionality, manufacturability, and assembly feasibility, identifying issues before builds and reducing rework
- Resolved assembly and fit issues during hands-on production builds of the MB-1, developing build instructions and coordinating with supply chain to address purchased part and raw material shortages and keep builds on schedule
- Drove critical-path items in project schedules to keep builds on track and reduce integration risk

Bajaj Auto Limited ***Jan 2021 - Jul 2024***

Mechanical Design Engineer, Powertrain Design *Pune, India*

- Owned end-to-end design of aluminum die cast (HPDC) components, injection-molded covers, sheet-metal brackets, and gaskets for ICE and EV powertrain platforms, from concept and detailed design through DFMEA, DVP, and high-volume production launch
- Directed CAD modeling and engineering drawing development in Siemens NX, reviewing designs for intent, tight tolerance control, and dimensional accuracy across HPDC, machined, and injection-molded components
- Applied FEA simulation in ANSYS across 20+ design studies, running static structural and contact pressure analyses to validate component integrity and sealing performance before physical prototyping
- Led multiple VAVE initiatives across powertrain components, including part rationalization, material substitution, and design optimization, cumulatively reducing component mass and delivering $15K+ in annual cost savings
- Automated gasket CAD modeling and validation workflows using NXOpen scripting, cutting design cycle time by 76%
- Drove root cause analysis for 100+ in-house and field failures, collaborating with quality, testing, and supply chain teams to implement corrective actions

TECHNICAL SKILLS

- **CAD & Simulation:** Siemens NX, SolidWorks, ANSYS Mechanical, FEA
- **Mechanical Design and Analysis**: GD&T, Tolerance Stack-up Analysis, DFM/DFA, DFMEA, Root Cause Analysis, DOE
- **Manufacturing and Prototyping**: Machining, 3D Printing, Die Casting, Injection Molding, Sheet Metal, HPDC
- **Programming:** MATLAB, Python
- **PLM and Tools:** Siemens Teamcenter

PROJECTS

Boeing Advanced Research Collaboration, University of Washington ***Jan 2025 – Dec 2025***

Autonomous Cleaning Robot, Mechanical Subsystem Integration *Seattle, WA*

- Led the mechanical architecture of an autonomous cleaning robot for an industrial sump environment, owning chassis design, component packaging, and enclosure development within tight height, weight, and wet-environment constraints
- Generated and evaluated multiple cleaning system concepts, down-selecting through a Pugh matrix to identify the most feasible solution for wet, low-maintenance operation on an inclined epoxy surface
- Developed CAD assemblies to validate fit, alignment, and interface compatibility, iterating on physical prototypes with material and component tradeoffs to balance durability, assembly simplicity, and field usability

EDUCATION

University of Washington, Seattle, WA *Expected Aug 2026*

Master of Science, Mechanical Engineering *GPA: 3.91*

National Institute of Technology, Trichy, India *Jun 2016 - Jun 2020*

Bachelor of Technology in Mechanical Engineering (Minor: Electrical and Electronics)

CERTIFICATIONS

Certified SolidWorks Professional (CSWP)